Post-Quantum Cryptography Migration

Part 14 - Where It Runs: Enterprise Applications & Delivery

Compiled by the Spaced Research Ledger

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The applications businesses actually live in, the ERP system, the video call, the remote desktop, are the last stop of the migration. Everything beneath them has to work first, and their vendors mostly move by inheriting the layers below.

This report covers three application families. Section 1 covers the ERP and business platforms, Section 2 collaboration and unified communications, and Section 3 virtual desktops and remote access.

The business platforms are further along than their reputation suggests. Oracle put the algorithms into the database itself, SAP ships quantum-safe TLS through its cryptographic library with a 2029 readiness target, and Workday published a dated migration program through 2028. The pattern is inheritance done deliberately, with published clocks.

Video went earliest and strangest. Zoom shipped post-quantum end-to-end encryption back in 2024, before some operating systems had the algorithms at all, while its competitors' end-to-end modes remain classical. And remote access shows the migration's rough edges plainly: assigning a post-quantum certificate to a Windows remote-desktop listener currently breaks the connection outright.

1. ERP & Business Platforms

The systems of record, ERP, HR, CRM, and ticketing, hold the data with the longest confidentiality requirements in any company. Their vendors control deep stacks, from database drivers to their own cryptographic libraries. This section covers who has shipped, who has published a clock, and who is still writing prerequisites.

Recent Developments

The version gates are now specific. Oracle's documentation confirms the post-quantum algorithms work only under its next-generation provider, in non-FIPS or FIPS 140-3 mode, never the legacy FIPS mode [1]. E-Business Suite certified on the 26ai database gains hybrid ML-KEM that older database lines lack [2]. The 19c line's release update 19.32 supports phased migration, new clients on TLS 1.3 while old ones stay behind [3].

SAP updated its Common Cryptographic Library with post-quantum capabilities for its product teams [4]. It supports post-quantum root CAs on its cloud platform, atop the quantum-safe TLS already in its ABAP systems [5]. Salesforce is retiring static RSA key exchange as a prerequisite step [6], and ServiceNow made a cryptographic-asset compliance product generally available, covering discovery through migration planning [7].

Current State

Oracle's database is the concrete anchor. Post-quantum support requires the new provider explicitly [1], covering ML-KEM for key exchange, ML-DSA for TLS signatures, and SLH-DSA for non-TLS signing only [1][1]. The FIPS-mode boundary is firm [1], the version gate is 19.32 [8], and the default remains the legacy provider, making everything an opt-in switch run by script [9][9]. FIPS 140-3 for the 19c line lands during 2026, retiring 3DES with it [9].

The strategy above the database is hybrid establishment across TLS, SSH, and IKEv2, designed to need no new certificates [10]. TLS 1.3 is named as the prerequisite [10], and hybrid mode arrived in the 26ai track [11].

SAP runs on a published clock. The company targets post-quantum readiness by 2029 across products and infrastructure [4]. Its cryptographic library already performs hybrid ML-KEM key agreement for TLS 1.3 on NetWeaver systems [12], with a FIPS 140-3 certified kernel [12].

The SaaS vendors differ mostly in specificity. Workday published the most detailed program: hybrid deployment beginning in 2026, full migration targeted by 2028 [13]. The stages run from discovery through hybrid TLS with anti-rollback, then internal federation and code signing, then automated agility [13]. Salesforce documents its harvest-now-decrypt-later posture [14], scoped to named service brands rather than a uniform platform timeline [14]. ServiceNow shipped a quantum-resistant key-management architecture that avoids asymmetric algorithms entirely [15].

2. Collaboration & Unified Comms

Every video call is encrypted twice over: hop by hop to the vendor's servers, and, when offered, end to end between participants. The second kind is where quantum matters most, because a recorded call is a harvest-now-decrypt-later target. This section covers the one vendor that moved and the standard machinery everyone else still runs.

Recent Developments

Webex joined the post-quantum tier. Its TLS 1.3 now supports a hybrid ML-KEM key agreement, and the same mechanism is integrated into the group-messaging security layer of its zero-trust end-to-end encrypted meetings [16].

Current State

Zoom holds the head start. It made post-quantum end-to-end encryption globally available for meetings in May 2024, using Kyber-768, claiming a first among communications providers [17]. The parameter targets security roughly equivalent to AES-192 [18], and coverage extended beyond meetings afterward [17]. The design keeps keys off Zoom's servers [19], building on classical end-to-end encryption shipped years earlier [17].

The rest of the market runs classical end-to-end modes on strong plumbing. Teams derives one-to-one call keys over DTLS, opaque to Microsoft [20], verified by spoken security codes [21], with everything baseline-encrypted regardless [20]. Its meeting mode negotiates group keys over TLS 1.3 with premium licensing and feature tradeoffs, recording and captions among them [20][20], and its signaling remains TLS 1.2 [22]. Google Meet encrypts in transit by default without client-side end-to-end [23][24]. Its client-side encryption lets organizations hold the keys at the cost of server-processed features [23][25], extended to meeting-room hardware [26] and, separately, to personal accounts [27].

Webex encrypts standard meetings hop by hop, its servers holding keys for interoperability [28]. Its zero-trust mode builds keys on participants' devices, downgrading transparently when a legacy endpoint joins [29][16]. Its calling upgrades to end-to-end opportunistically [30], the app rides TLS and strong media ciphers [31], and its end-to-end lineage dates to 2008 [32]. Interoperating devices negotiate the standard cipher menu [28].

The underlying standards are the fixed layer. The DTLS extension keying secure media dates to 2010 and remains the baseline [33], scoped to two-party sessions [34], implemented by Meet [23] and mandated by WebRTC for browser media [35]. The companion framework carries certificate fingerprints through call signaling [36], protectable by identity signatures [33]. Secure media transport itself is symmetric [37], with session keys exchanged over the TLS signaling channel [37], required end to end on certified trunks [35]. All of it inherits its quantum future from the TLS beneath.

3. VDI & Remote Access

Remote desktops and virtual-desktop gateways carry keystrokes, screens, and credentials across the internet, a harvest target as good as any. Their post-quantum story is the Windows TLS story plus a delivery-controller race, with one sharp incompatibility already on record. This section covers what works, what breaks, and what to configure.

Recent Developments

The protocol layer arrived with the July Windows update. Hybrid post-quantum key exchange shipped into Schannel for Windows 11 and Windows Server 2025, moving from developer API to protocol [38]. Three hybrid groups pair ML-KEM with classical curves [38].

Current State

Citrix's delivery controller leads the category. NetScaler ships production hybrid post-quantum key exchange, browser-compatible [39], first as an early-access preview in April 2025, then generally available that August [40][40]. It sits on the defense procurement list [40], released alongside the virtual-desktop suite's new long-term cadence [41], and is framed squarely against harvest-now-decrypt-later [41]. The interoperability warning comes from a competitor's stack: Omnissa Horizon's gateway times out when a current Safari forces post-quantum key negotiation the landing page cannot handle [42].

Windows remote access is configurable but sharp-edged. None of the new hybrid groups activate automatically, and TLS 1.2 environments get nothing [38]. Enablement runs through Group Policy's curve ordering [43], atop the API primitives shipped in November 2025 [44]. The certificate side lags dangerously.

Assigning an ML-DSA certificate to a remote-desktop listener currently makes clients fail to connect outright [45], and the broader certificate support is explicitly limited beyond evaluation [45]. Self-hosted gateways can enable the hybrids once clients are confirmed capable [43], with the rollout path documented since the Insider previews [46].

The category guidance is consistent. Hybrid post-quantum in SSH and IPsec, the protocols under VPN-based access, keeps gaining operational adoption [47]. Migration frameworks tell enterprises to inventory VPN and remote-access key exchange as its own category [48] and to prioritize remote-access systems in roadmaps [49]. They warn of long hybrid coexistence at federation boundaries where partners migrate on their own clocks [50]. And they advise reading cloud responsibility models carefully, since some protections arrive transparently and others need opting in [51].

About This Report

The Spaced Research Ledger is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from four to seven systems across competing labs, so no single model both finds a claim and judges it. The Spaced Research Ledger is designed and maintained by Ridley DiSiena.

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